

Homework 4 - Writeup

Team members

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1. Consider a paging system with the page table stored in memory. (a) If a memory reference takes 50 nanoseconds, how long does a paged memory reference take? (b) If we add TLBs, and if 75 percent of all page-table references are found in the TLBs, what is the effective memory reference time? (Assume that finding a page-table entry in the TLB takes 2 nanoseconds, if the entry is present.)

Answer:

- (A) If a memory reference takes 50 nanoseconds, a paged memory reference will take a total of 100ns.
- (B) If we add TLBs, the memory reference time will become 64.5ns.
- Computation work:

$$\begin{aligned} \text{A) Time} &= \text{page table lookup time} + \text{memory access time} \\ &= 50\text{ns} + 50\text{ns} \\ &= 100\text{ ns} \end{aligned}$$

$$\begin{aligned} \text{B) entry found:} & \text{TLB lookup time} + \text{memory access time} \\ &= 2 + 50 \\ &= 52\text{ns} \end{aligned}$$

$$\begin{aligned} \text{entry miss} &= \text{TLB lookup time} + \text{page lookup time} + \text{memory access time} \\ &= 2 + 50 + 50 \\ &= 102\text{ ns} \end{aligned}$$

$$\begin{aligned} \text{average} &= (\text{hit rate} \times \text{hit time}) + (\text{miss rate} \times \text{miss time}) \\ &= (0.75 \times 52) + (0.25 \times 102) \\ &= 39 + 25.5 \\ &= 64.5\text{ ns} \end{aligned}$$

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2. Consider a computer system with a 32-bit logical address and 8-KB page size. The system supports up to 1 GB of physical memory. How many entries are there in a conventional, single-level page table?

Answer:

- The amount of entries would equal 524,288
- Computation work:

$$\begin{aligned} 2) \# \text{ of pages} &= \frac{\text{logical address space size}}{\text{page size}} \\ &= \frac{2^{32} \text{ bytes}}{2^{13} \text{ bytes per page}} \\ &= 2^{32-13} \\ &= 2^{19} = 524,288 \end{aligned}$$

3. The following is a page table for a system with 12-bit virtual and physical addresses and 256-byte pages. Free page frames are to be allocated in the order 9, F, D. A dash for a page frame indicates that the page is not in memory. Convert the following virtual addresses to their equivalent physical addresses in hexadecimal. All numbers are given in hexadecimal. In the case of a page fault, you must use one of the free frames to update the page table and resolve the logical address to its corresponding physical address. (a) 0x2A1 (b) 0x4E6

Answer:

- (A) The physical address would be 0x9A1
- (B) The physical address would be 0xFE6
- Computation work:

A) $0x2A1 \rightarrow$ convert to binary $= 0010\ 1010\ 0001_2$
page # = $0x2$, offset = $0xA1$

$$\begin{aligned}\text{physical address} &= \text{frame \#} \times \text{page size} + \text{offset} \\ &= (9 \times 256) + 0xA1 \\ &= 0x900 + 0xA1 \\ &= 0x9A1\end{aligned}$$

B) $0x4E6 \rightarrow$ convert to binary $\rightarrow 0100\ 1110\ 0110_2$
page # = $0x4$ offset = $0xE6$

$$\begin{aligned}\text{physical address} &= \text{frame \#} \times \text{page size} + \text{offset} \\ &= (F \times 256) + 0xE6 \\ &= 0xF00 + 0xE6 \\ &= 0xFE6\end{aligned}$$

4. Given a system where the page size is 4Kbytes and given two variables x and y who are placed in memory at addresses $0x1230$ and $0x2340$ respectively, which statement(s) are true? (a) The physical address of y is larger than the physical address of x (b) We cannot tell which physical address is larger until we translate the addresses (c) It depends on whether we have a TLB hit or not

Answer:

- ☐ (A) True
 - ☐ (B) False
 - ☐ (C) False
5. Consider a system where virtual memory address size is 38 bits and physical address size is 30 bits. The memory is paged with a total of 2^{16} frames in physical memory. (a) What is the page size? (b) What is the size of a flat page table assuming that the Page table entry is 8 bytes? (c) If we use a two level page table, what is the size of the outer (or first level) page table?

Answer:

- ☐ (A) 16 KB $\rightarrow 2^{14}$ bytes
- ☐ (B) 128 MB $\rightarrow 2^{27}$ bytes
- ☐ (C) 64 KB $\rightarrow 2^{16}$ bytes

- Computation work:

A) offset bits = $30 - 16 = 14$ bits
page size = 2^{14} bytes = 16 KB

B) # of virtual pages = $38 - 14 = 24$

total # of pages = 2^{24}
each page is 8 bytes
total page table size = $2^{24} \times 8$
 $= 2^{27}$ bytes = 128 MB

C) inner page table = $\frac{\text{page size}}{\text{PTE Size}} = \frac{2^{14}}{2^3} = 2^{11}$ entries

outer page table = $24 - 11 = 13$ bits
 $= 2^{13} \times 8$
 $= 2^{16} = 64$ KB

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